

Codebook

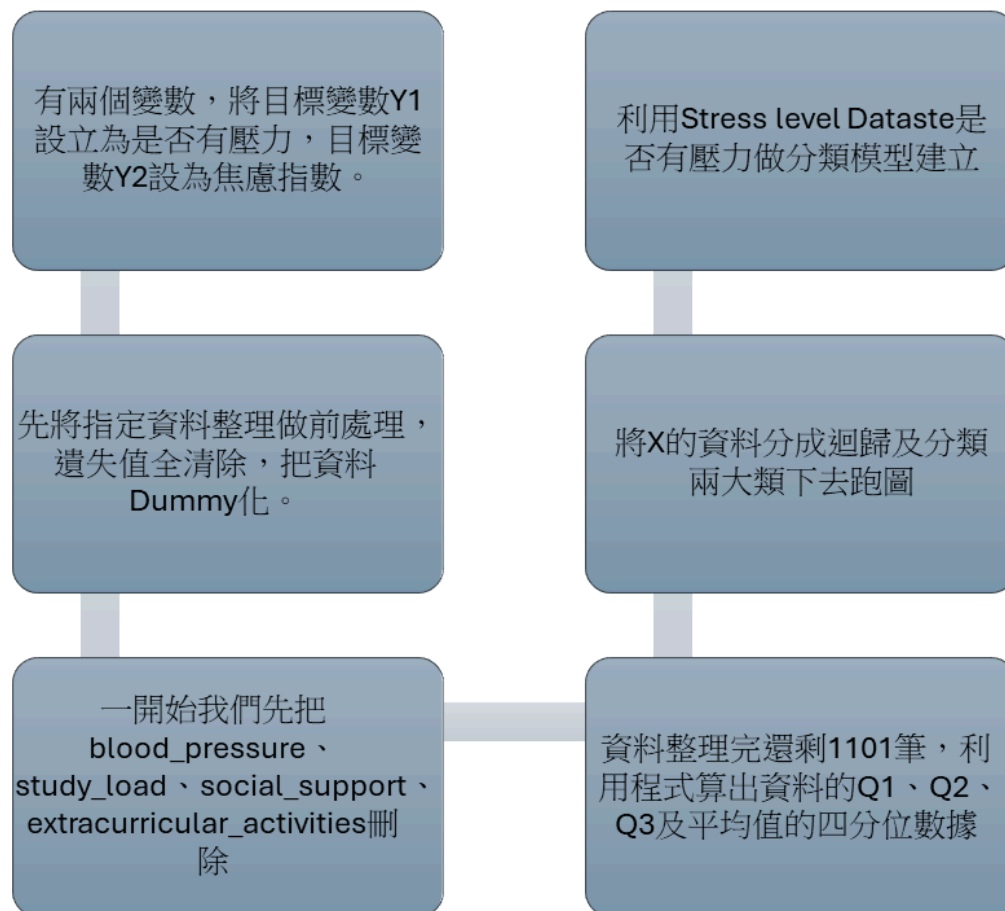
1. Date: 2024/1/14

2. Name of the data file: **D-StressLevelDataset.CSV**

3. Data overview:

Total sample size	798
Independent variables	
Total of numeric variables	4
Total of categorical variables	4
Sum	8
Dependent variables	
Total of numeric variables	0
Total of categorical variables	1

4.



5. Variables overview:

Index	Variable	Definitio n	Code	Data type
Y	stress	Whether it's stressful or not	0: No 1: Yes	Category
X1	anxiety	anxiety level	0-20	Numeric
X2	health history	With or without	0: No 1: Yes	Category

		medical records		
X3	self esteem	affect self-esteem	0-20	Numeric
X4	depression	level of frustration	0-20	Numeric
X5	headache	headache index	0、1、2、3、4、5	Category
X6	sleep quality	Sleep quality level	0、1、2、3、4、5	Category
X7	breathing problem	respiratory problem index	0、1、2、3、4、5	Category
X8	noise level	Noise impact level	0、1、2、3、4、5	Category
X9	living conditions	survival index	0、1、2、3、4、5	Category
X10	safety	security index	0、1、2、3、4、5	Category

X11	basic needs	basic needs index	0、1、2、3、4、5	Category
X12	academic performance	academic impact index	0、1、2、3、4、5	Category
X13	teacher-student relationship	teacher-student relationship index	0、1、2、3、4、5	Category
X14	future career concerns	Worry about future career impact index	0、1、2、3、4、5	Category
X15	peer pressure	peer pressure index	0、1、2、3、4、5	Category
X16	bullying	Situation of being bullied	0、1、2、3、4、5	Category

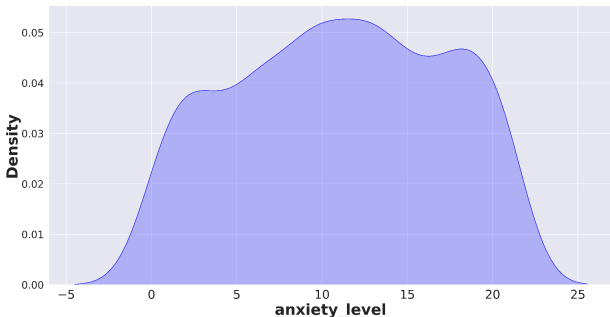
6. Variable description :

Dependent variable(known as "y" or output)

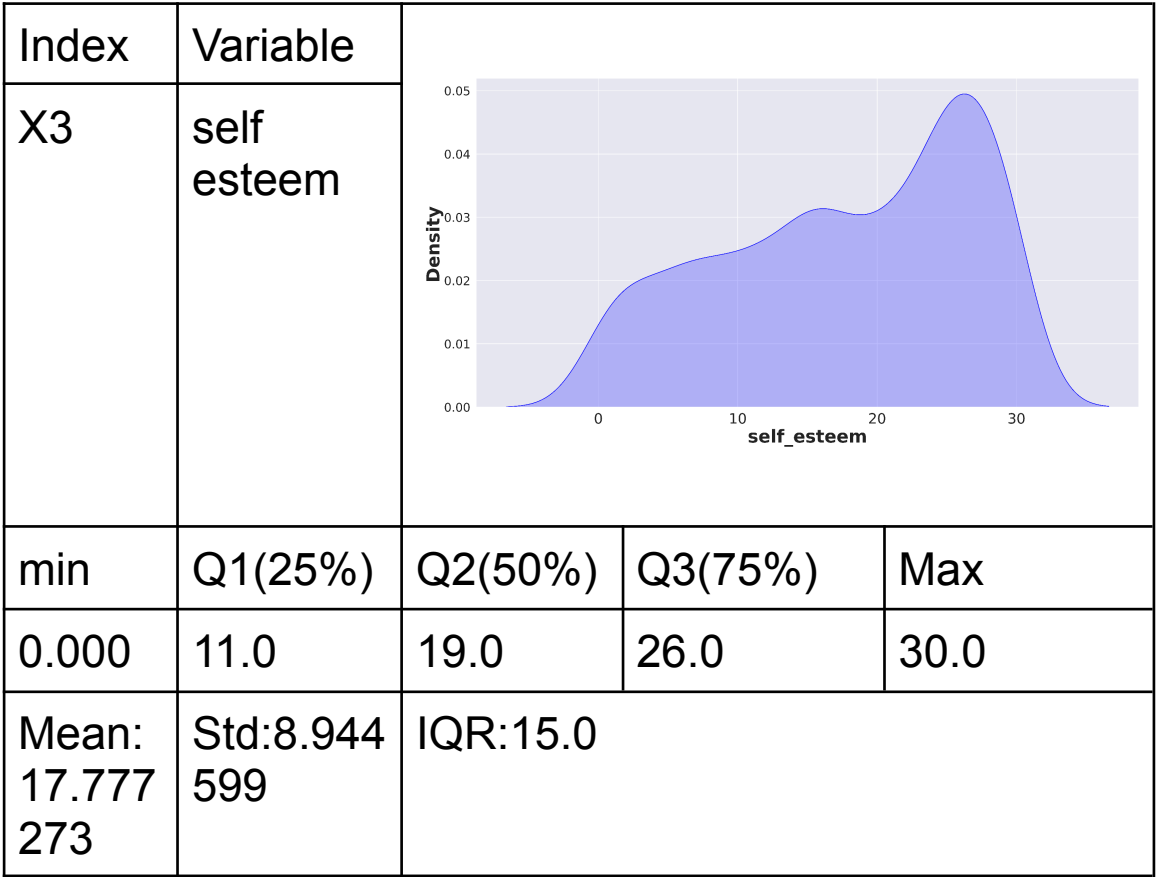
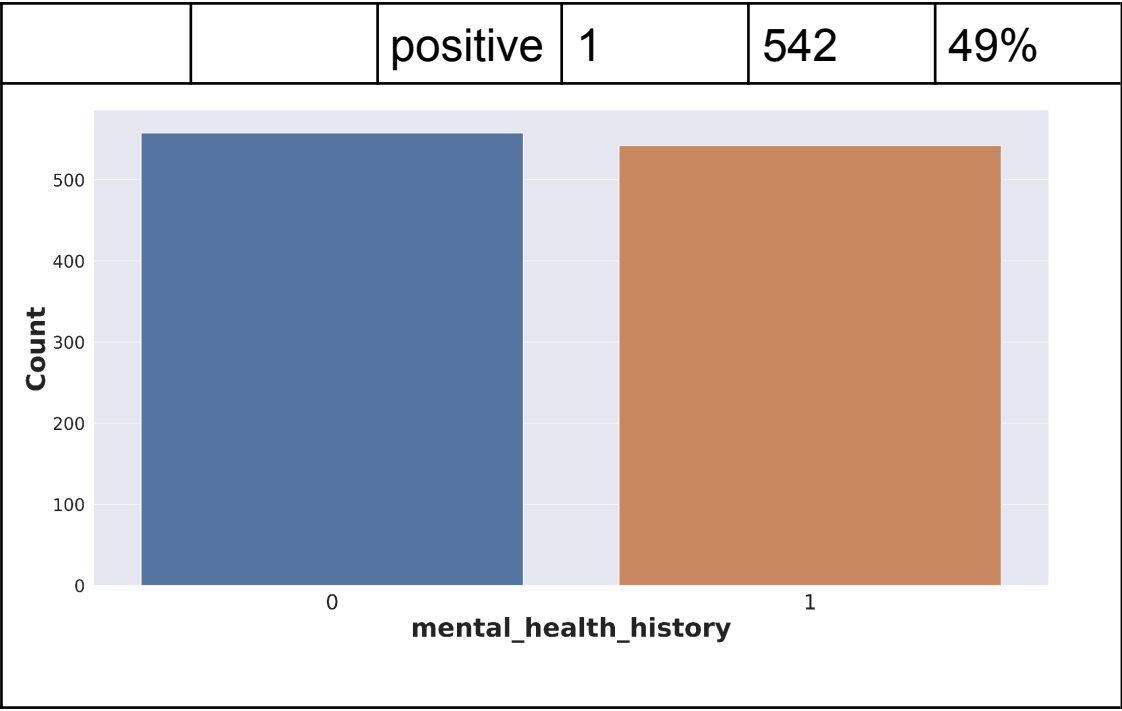
Index	Variable	Description	Class code	Count	%
-------	----------	-------------	------------	-------	---

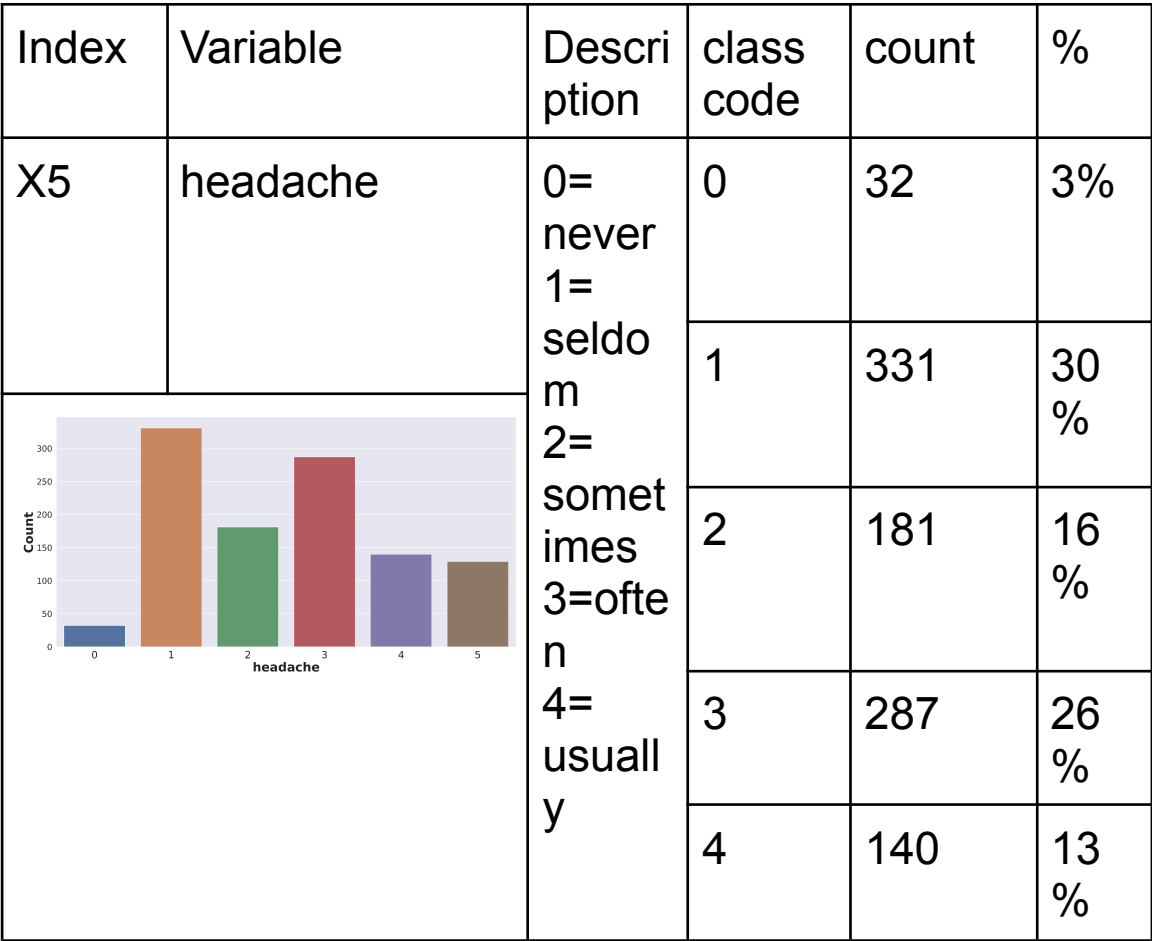
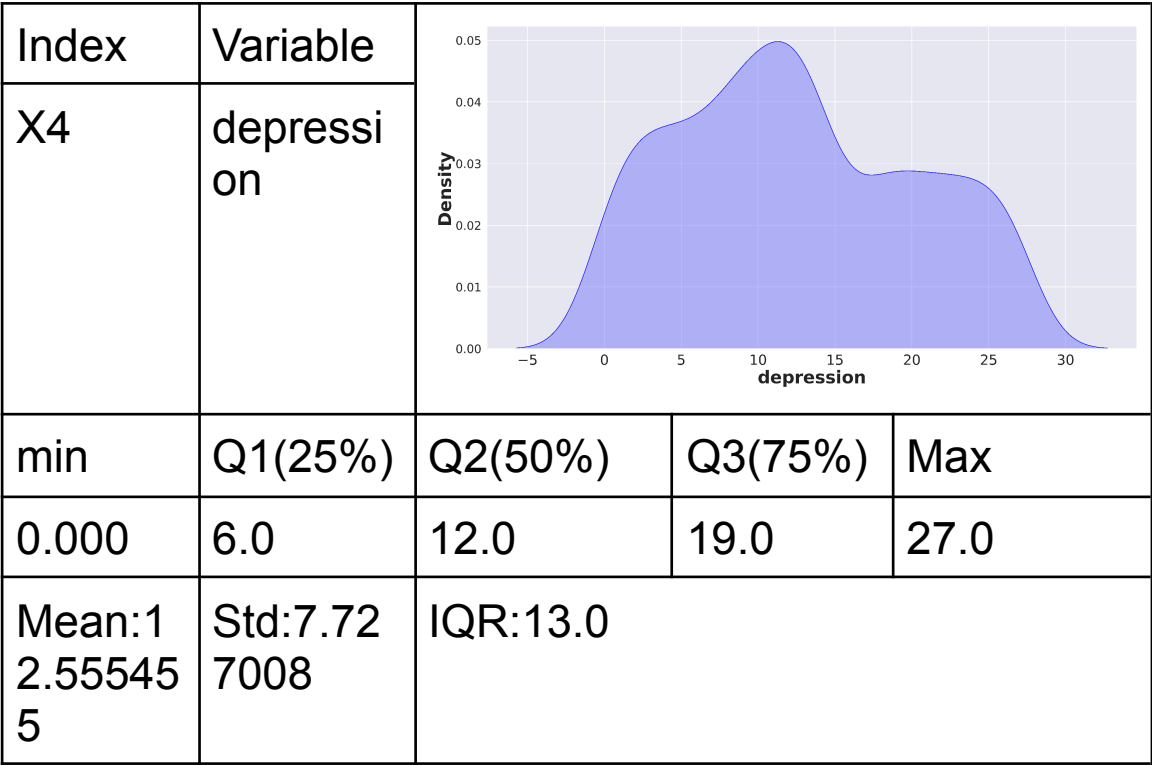
Y	stress	negative	0	482	60
		positive	1	309	40
Note					

Independent variable(known as“X”or input)

Index	Variable			
X1	anxiety			
min	Q1(25%)	Q2(50%)	Q3(75%)	Max
0.000	6.000	11.000	16.000	21.000
Mean:1 1.06363 6	Std:6.11 7558	IQR:10.000		

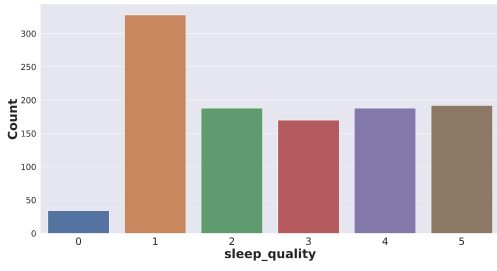
Index	Variable	Description	class code	count	%
X2	health history	negative	0	558	51%





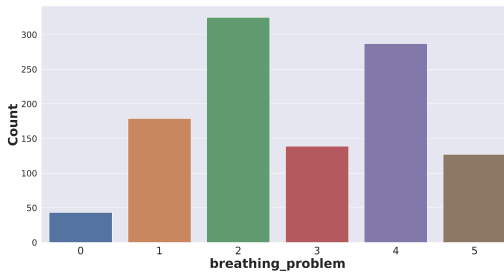
	5= always	5	129	12 %
--	--------------	---	-----	---------

Index	Variable	Description	class code	count	%
X6	sleep quality	0= never 1= seldom 2= sometimes 3=often 4= usually 5= always	0	34	3%
			1	328	30 %
			2	188	17 %
			3	170	15 %
			4	188	17 %
			5	192	18 %

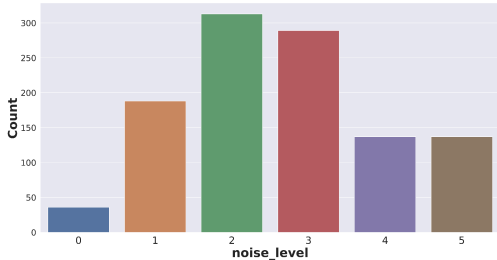


sleep_quality	Count
0	34
1	328
2	188
3	170
4	188
5	192

Index	Variable	Description	class code	count	%
X7	breathing problem	0= never 1= seldom 2= sometimes 3= often 4= usually 5= always	0	43	4%
			1	179	16%
			2	325	29%
			3	139	13%
			4	287	26%
			5	127	12%

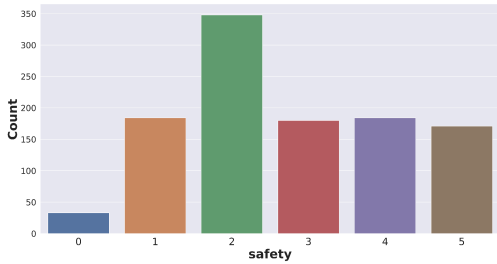


Index	Variable	Description	class code	count	%
X8	noise level	0= never 1= seldom 2= sometimes	0	38	3%
			1	188	17%

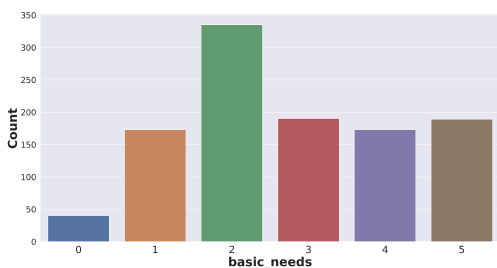
	3=often 4=usually 5=always			
		2	313	28%
		3	289	26%
		4	137	13%
		5	137	13%

Index	Variable	Description	class code	count	%
X9	living conditions	0=never 1=seldom 2=sometimes 3=often 4=usually 5=always	0	29	3%
			1	176	16%
			2	346	31%
			3	327	30%
			4	189	17%
			5	33	3%

Index	Variable	Description	class code	count	%
X10	safety	0= never 1= seldom 2= sometimes 3= often 4= usually 5= always	0	33	3%
			1	184	17%
			2	348	32%
			3	180	16%
			4	184	17%
			5	171	16%

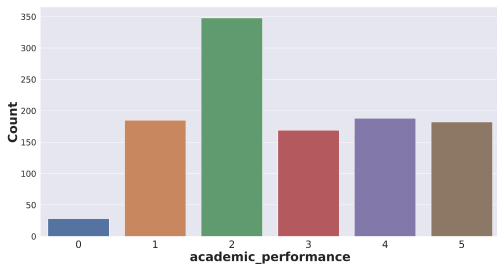


Index	Variable	Description	class code	count	%
X11	basic needs	0= never 1= seldom 2= sometimes 3= often	0	40	4%
			1	173	16%
			2	335	30%

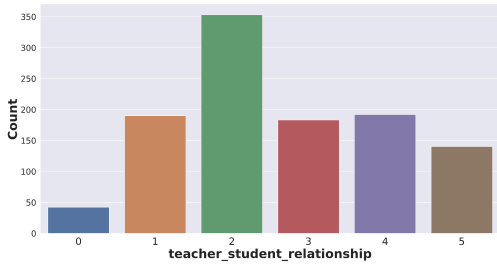


	4= usually 5= always	3	190	17 %
		4	173	16 %
		5	189	17 %

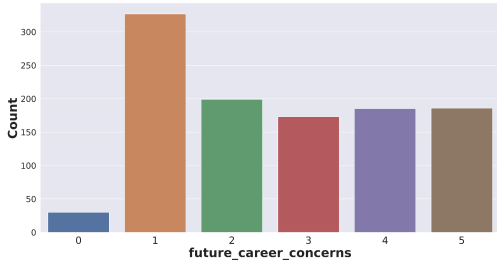
Index	Variable	Description	class code	count	%
X12	academic performance	0= never 1= seldom 2= sometimes 3=often 4= usually 5= always	0	28	3%
			1	185	17 %
			2	348	32 %
			3	169	15 %
			4	188	17 %
			5	182	17 %



Index	Variable	Description	class code	count	%
X13	teacher student relationship	0= never 1= seldom 2= sometimes 3= often 4= usually 5= always	0	42	4%
			1	190	17%
			2	353	32%
			3	183	17%
			4	192	17%
			5	140	13%



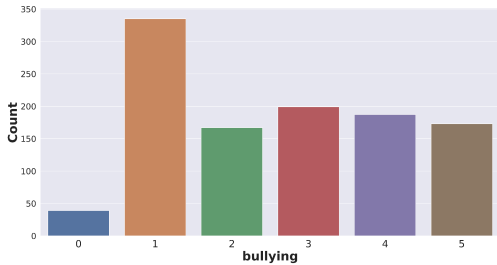
Index	Variable	Description	class code	count	%
X14	future career concerns	0= never 1= seldom 2= sometimes	0	30	3%
			1	327	30%

	3=often			
	4=usually	2	199	18%
	5=always	3	173	16%
		4	185	17%
		5	186	17%

Index	Variable	Description	class code	count	%
X15	peer pressure	0=never 1=seldom 2=sometimes 3=often 4=usually 5=always	0	37	3%
			1	181	16%
			2	355	32%
			3	172	16%
			4	174	16%

		5	181	16 %
--	--	---	-----	------

Index	Variable	Description	class code	count	%
X16	bullying	0= never 1= seldom 2= sometimes 3=often 4= usually 5= always	0	39	4%
			1	335	30 %
			2	167	15 %
			3	199	18 %
			4	187	17 %
			5	173	16 %



6.Odds ratio

	Negative	Positive	Pos_odd	Pos_OR
mental_health_history				
0	335	223	0.665672	1.000000
1	38	504	13.263158	19.924475

	Negative	Positive	Pos_odd	Pos_OR
sleep_quality				
0	13	21	1.615385	1.000000
1	8	320	40.000000	24.761905
2	13	175	13.461538	8.333333
3	14	156	11.142857	6.897959
4	162	26	0.160494	0.099353
5	163	29	0.177914	0.110137

	Negative	Positive	Pos_odd	Pos_OR
peer_pressure				
0	17	20	1.176471	1.000000
1	151	30	0.198675	0.168874
2	173	182	1.052023	0.894220
3	10	162	16.200000	13.770000
4	12	162	13.500000	11.475000
5	10	171	17.100000	14.535000

	Negative	Positive	Pos_odd	Pos_OR
bullying				
0	15	24	1.600000	1.000000
1	312	23	0.073718	0.046074
2	14	153	10.928571	6.830357
3	13	186	14.307692	8.942308
4	6	181	30.166667	18.854167
5	13	160	12.307692	7.692308

7.決策樹建模(分類)

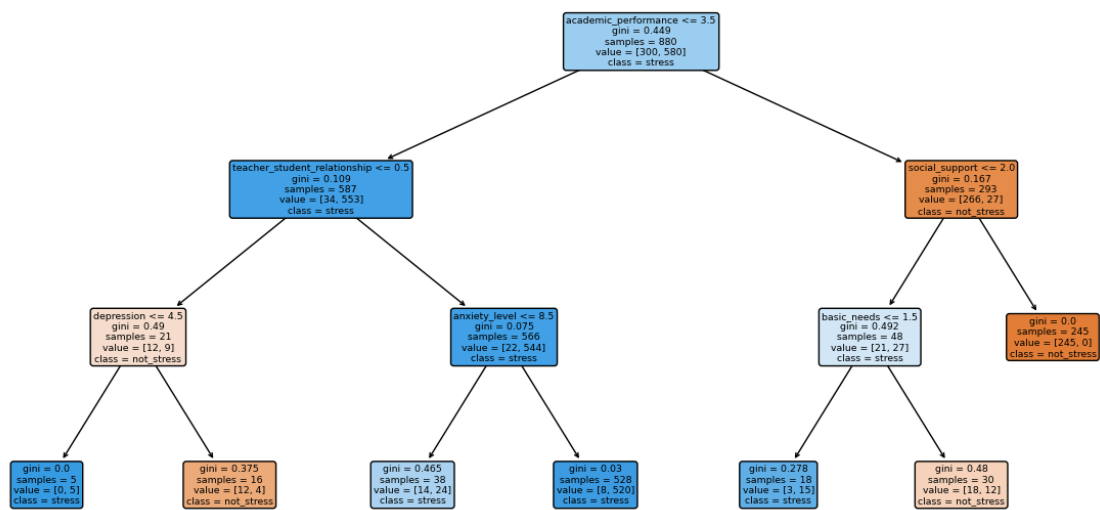
random state:205



CART

```
=====
Accuracy:      87.73
Bal ACC:       89.09
Precision:     96.15
F1 score:      90.25
Sensitivity:   85.03
Specificity:   93.15
AUC:           94.67
ThresHold:     0.9848484848484849
+-----+
```

```
[[ 68   5]
 [ 22 125]]
```

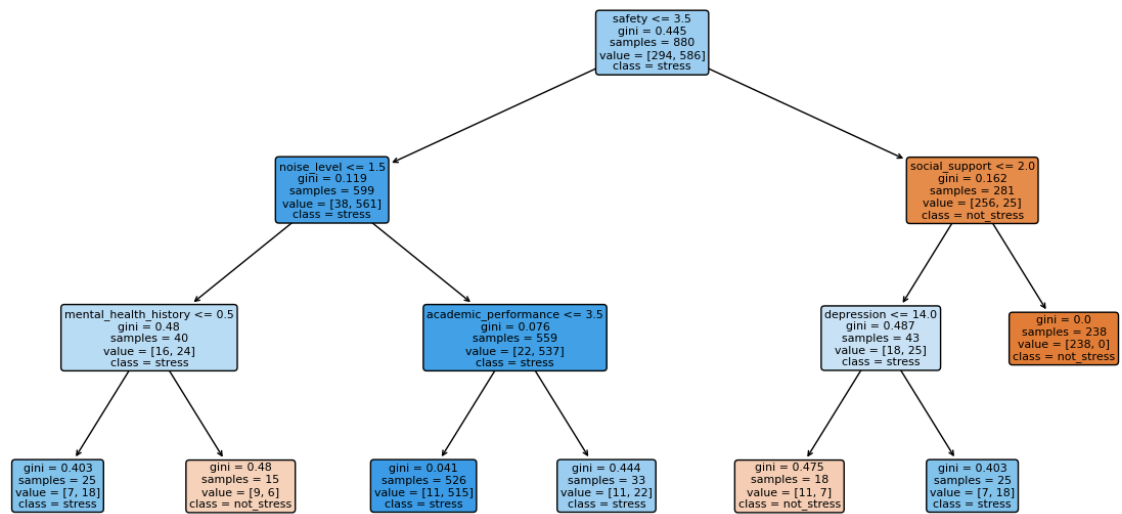


random state:643

CART

```
=====
Accuracy:      91.82
Bal ACC:       92.5
Precision:     96.95
F1 score:      93.38
Sensitivity:   90.07
Specificity:   94.94
AUC:           96.36
ThresHold:     0.9790874524714829
+-----+
```

```
[[ 75   4]
 [ 14 127]]
```

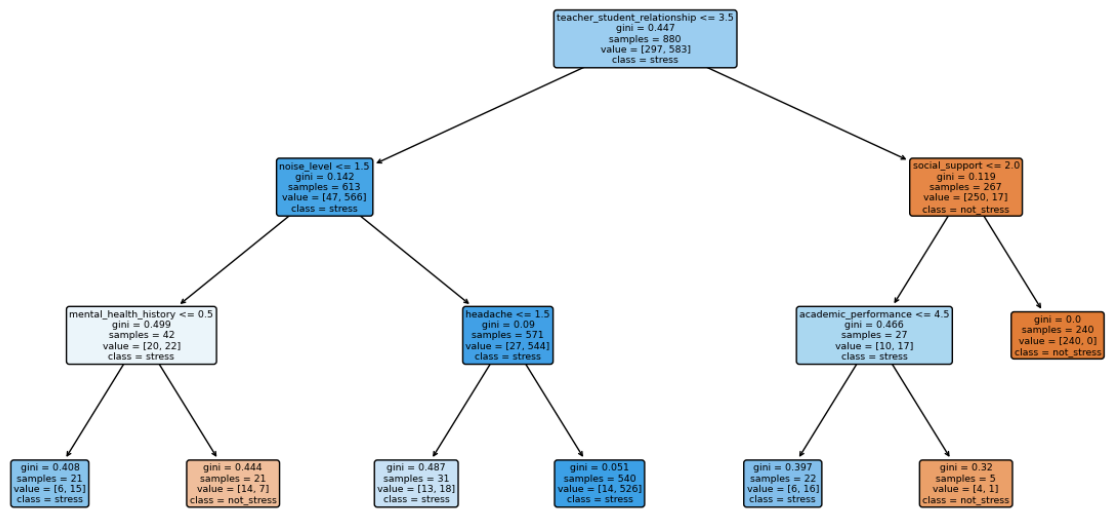


random state:866

➡ CART

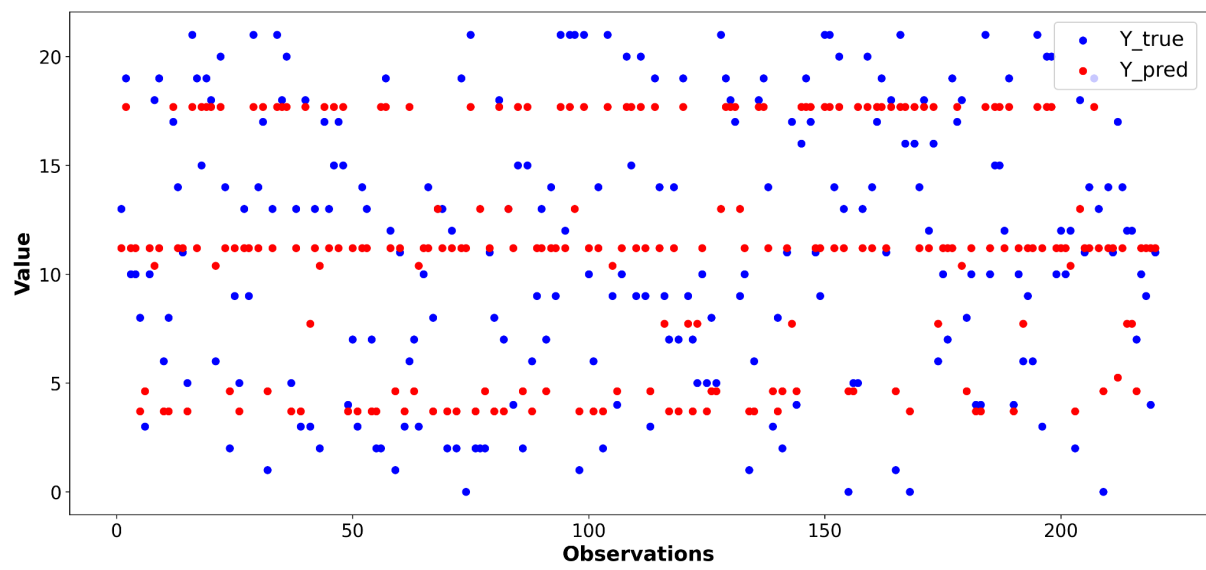
```

=====
Accuracy:      91.36
Bal ACC:       90.61
Precision:     93.71
F1 score:      93.38
Sensitivity:   93.06
Specificity:   88.16
AUC:           93.74
ThresHold:     0.7272727272727273
+-----+
[[ 67   9]
 [ 10 134]]
  
```

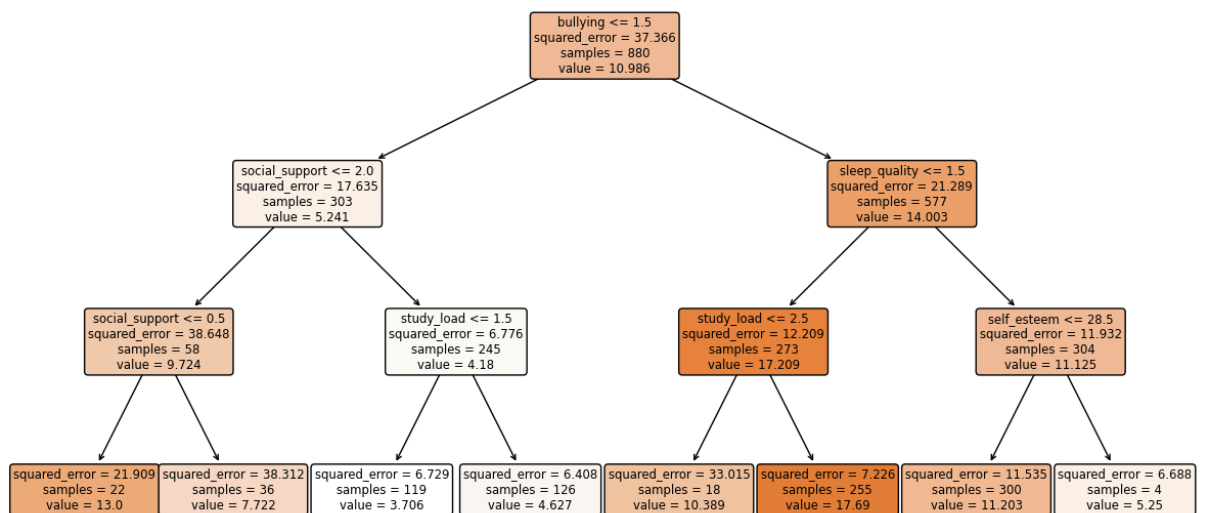
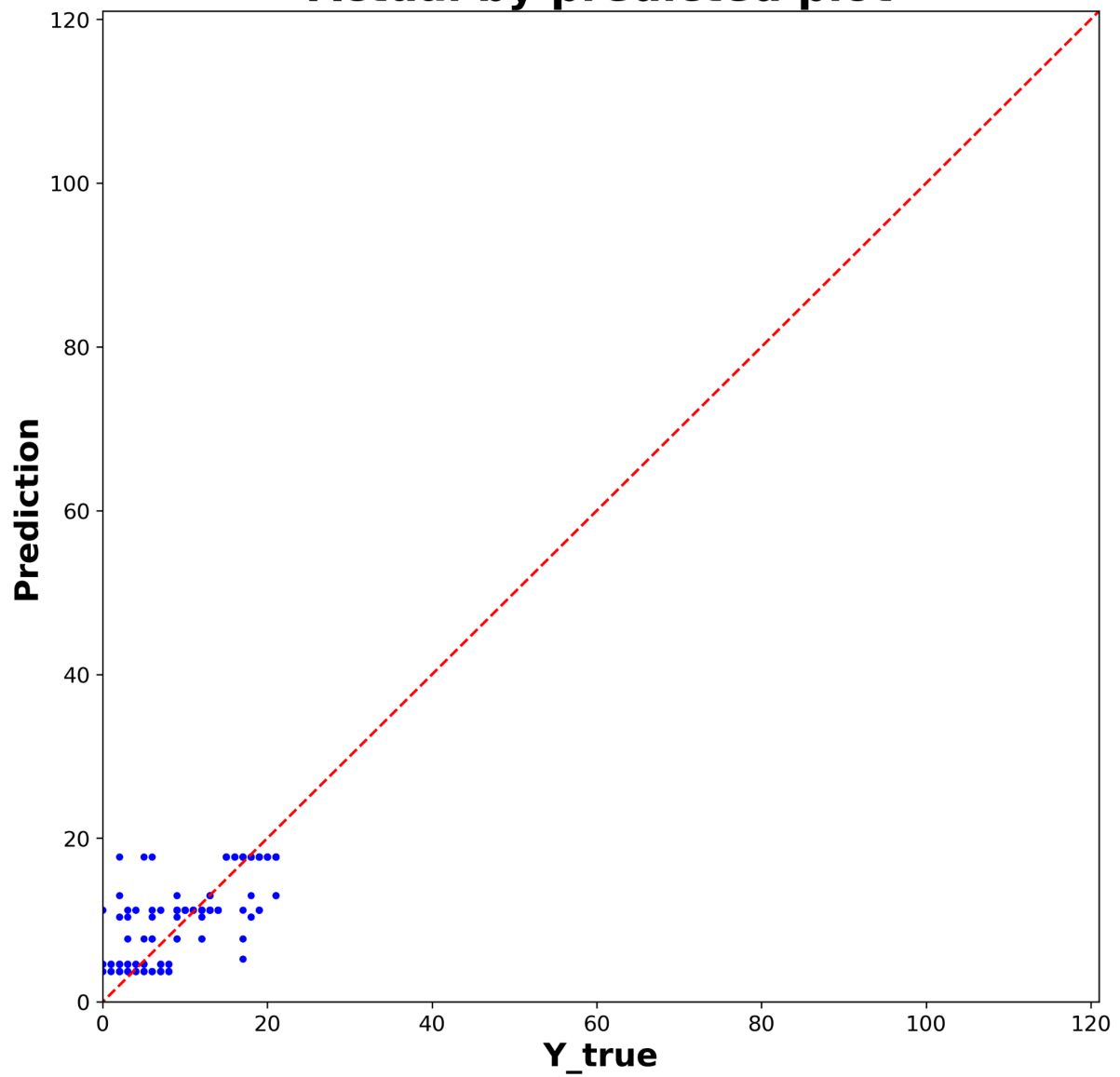


決策樹建模(連續變數)

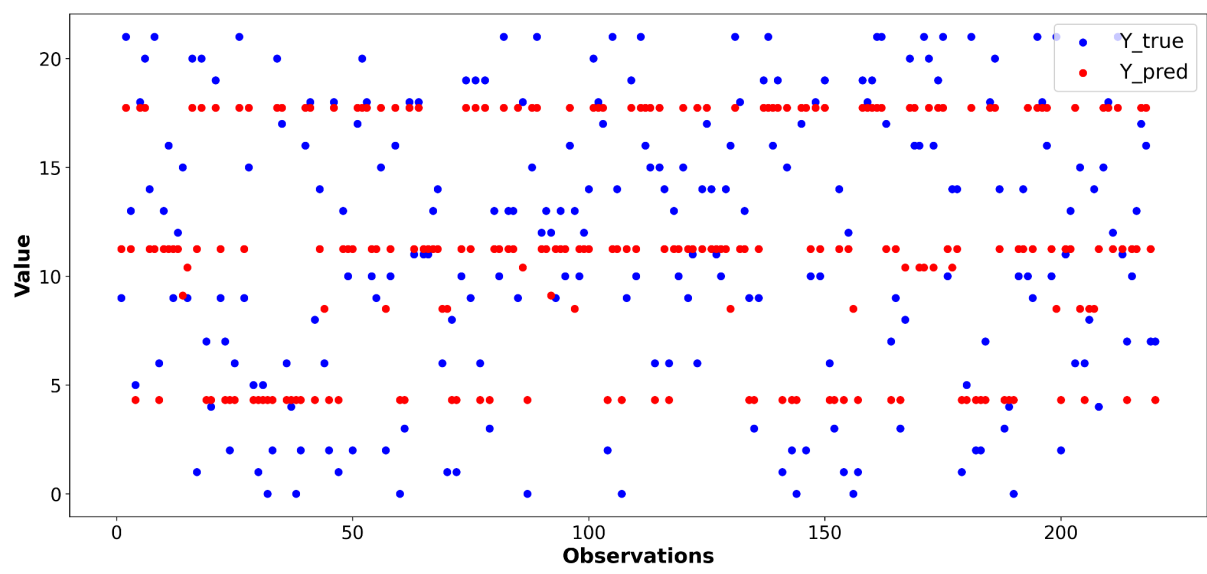
random state:229



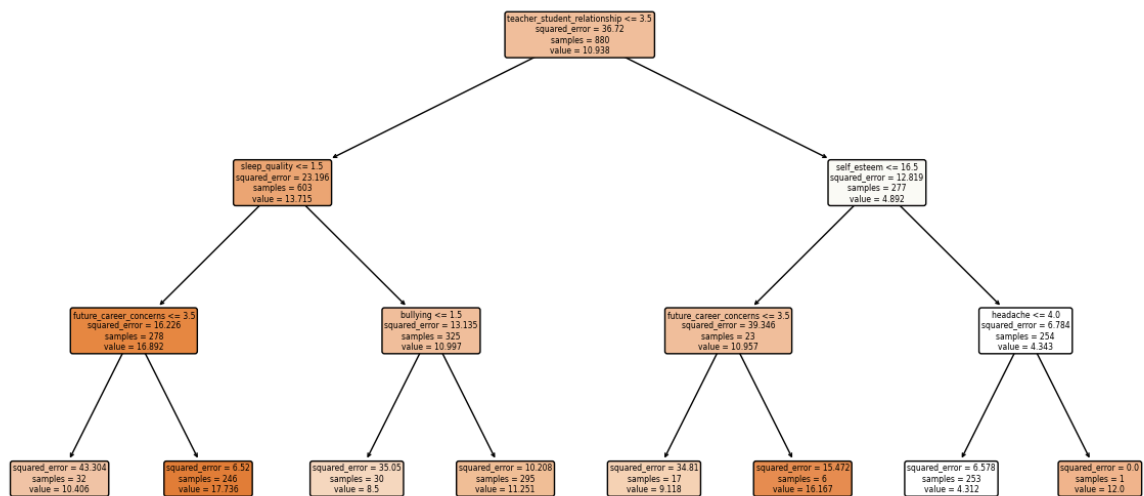
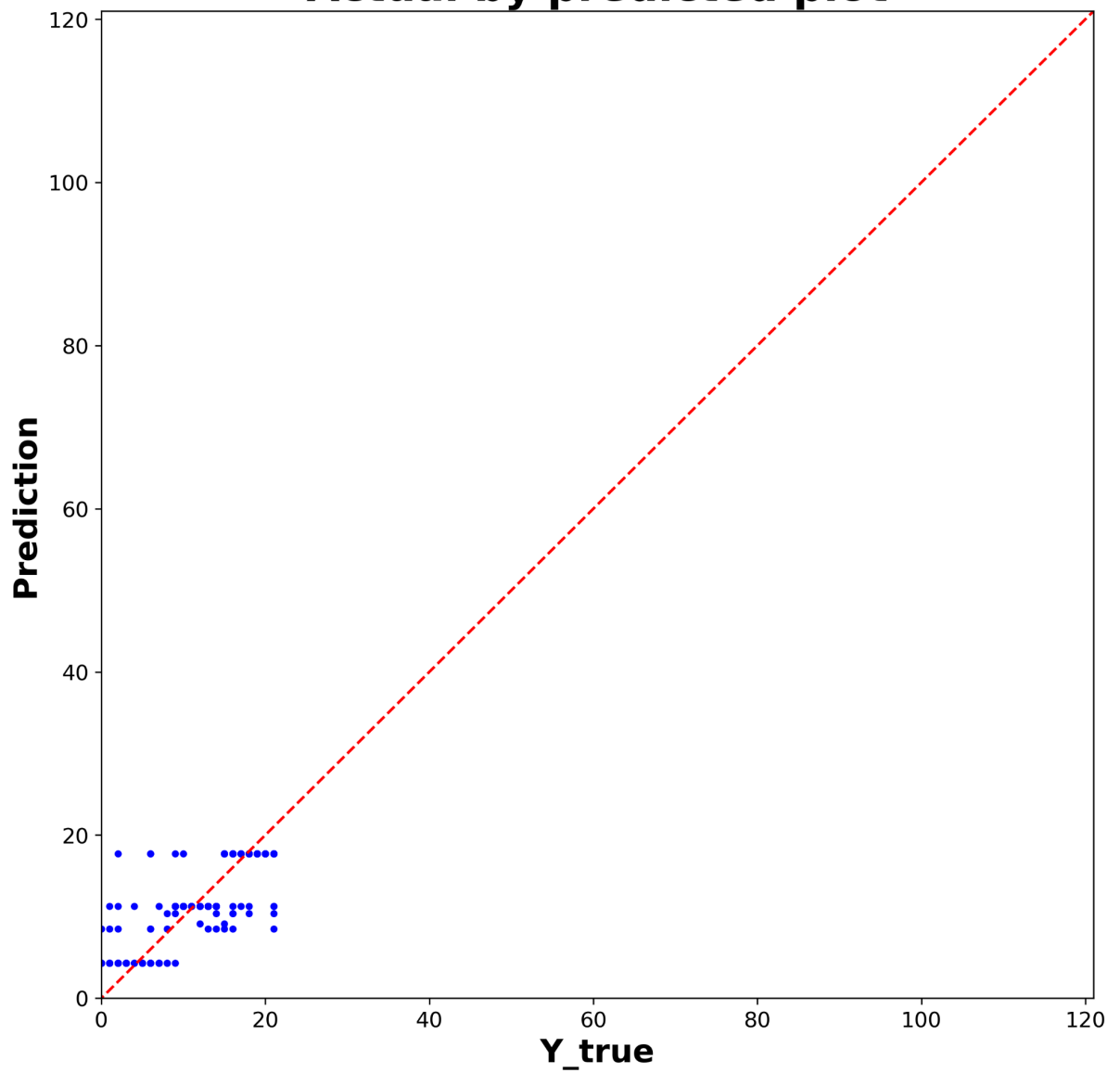
Actual by predicted plot



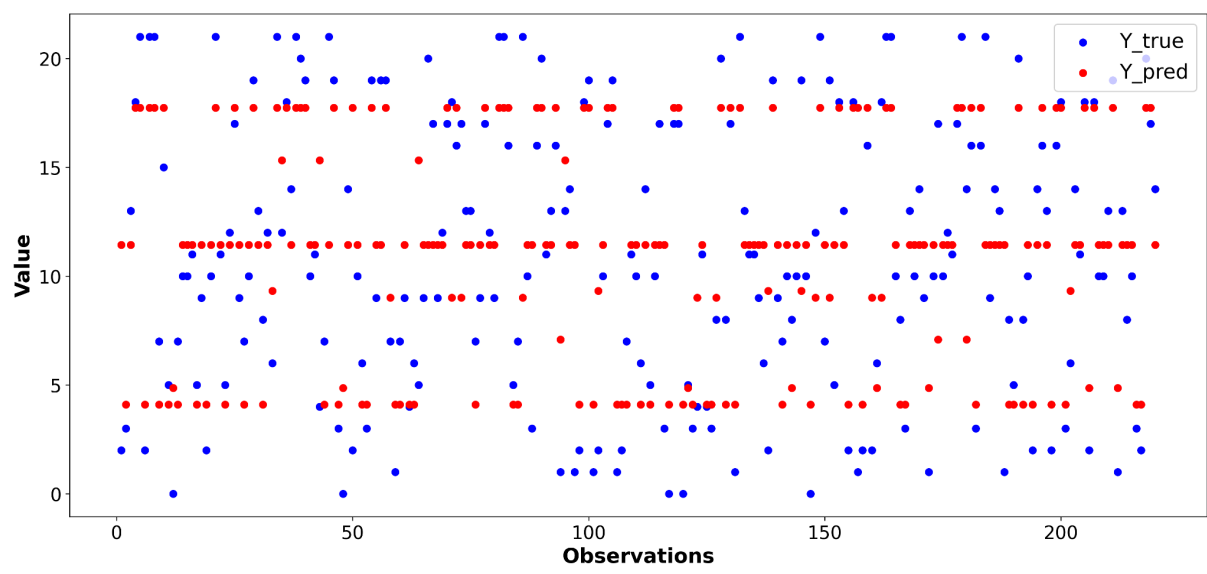
radom state:915



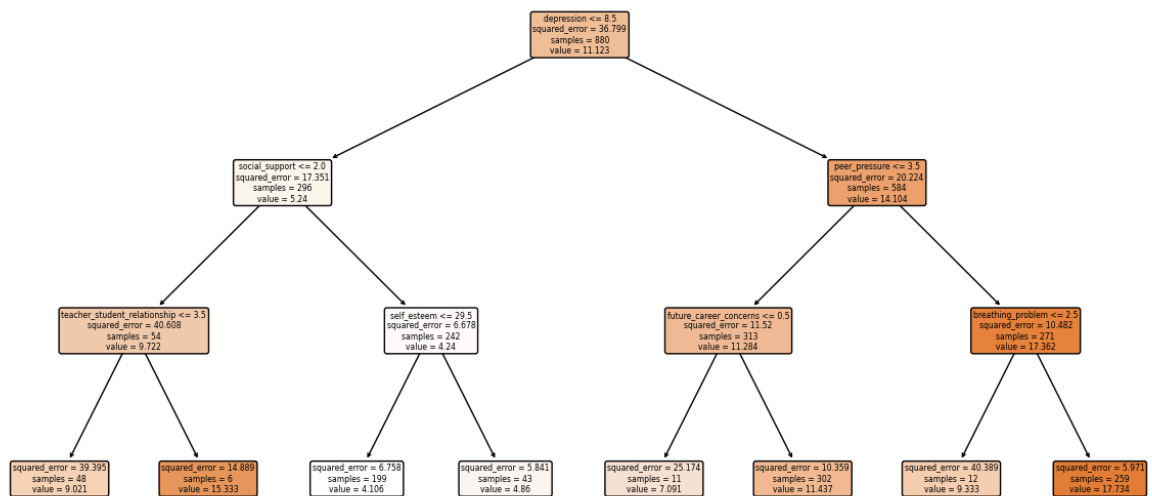
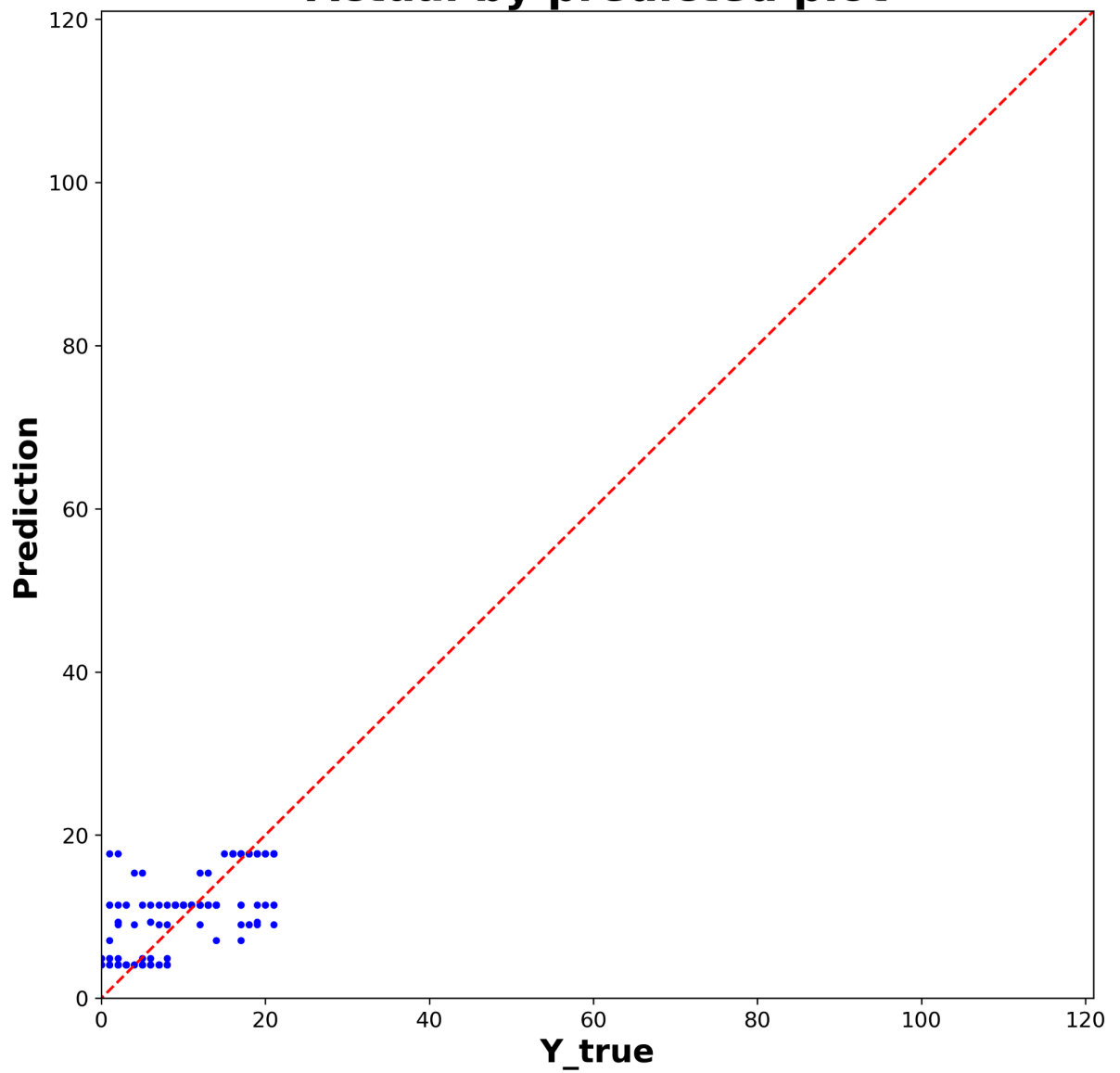
Actual by predicted plot



radom state:1003



Actual by predicted plot



分工表

資料探勘. 期末報告.

1. 30% 陳若儀.

2. 30% 張莉詠.

3. 30% 鍾心柔.

4. 10% 林沛茹.